

Masters Eligibility Analytics Dashboard

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Introduction

In the realm of higher education, efficiently assessing candidate eligibility is crucial for program admissions. This assignment focuses on developing an interactive Power BI dashboard that visualizes the eligibility status of candidates applying for the Master of Business Analytics program. Using a provided CSV data source, we will perform comprehensive data preparation, cleaning, and wrangling to ensure that the dataset meets the necessary standards for analysis. The dashboard will visualize critical eligibility criteria, including academic qualifications and professional experiences, thereby providing insights that facilitate informed decision-making for admissions. By utilizing various visualization techniques, such as pie charts, funnel charts, and stacked bar charts, the dashboard will enhance the user experience and allow stakeholders to explore candidate data dynamically. This project also emphasizes the importance of accurate DAX calculations to compute post-qualifying experience, which is a critical aspect of the evaluation process.

The overview on the data source

The provided CSV file primarily contains columns for ID, gender, academic qualifications, experience, and pay verification. In the "academic" column, the data includes details of the offered institute, degree, start date, end date, and class. In the “experiences” column, the data includes the company names, the positions held, the starting date and the ending date of the positions held.

Data preparation, cleaning & Data wrangling

Splitting Academic Qualifications

During the data preparation and cleaning process, we separated each academic qualification into new columns for each candidate by splitting the data at the "{" symbol. Any columns that contained only brackets after splitting were removed.

To eliminate the "institute:" prefix, we split the data at "institute:" and removed the resulting empty columns. We then further split the data at ",from" to separate the institute names. The starting date of each qualification was isolated by splitting the data at ",to" and the end date was extracted by splitting at ",class" Finally, the class information was separated by splitting the data again at "}" and the unnecessary brackets were removed. This process was repeated for all academic qualifications.

Splitting Experience Data

Similarly, to separate the "experience" column for each candidate, we first split the data at the "{" symbol and removed any resulting empty columns. We then removed the "company:" prefix by splitting the data at "company:" and removed any empty columns. The company information was separated by splitting the data again at ",position" To extract the position held, we split the data at ",from" and to get the starting and ending dates, we split at ",to" and "}" respectively. We removed any unnecessary brackets in this process and repeated it for all job experiences.

Renaming Columns

We systematically renamed the columns to ensure clarity and consistency. For academic qualifications, the columns related to the institute names were renamed as “Institute 1”, “Institute 2”, and so forth, depending on how many qualifications a candidate had. Similarly, the columns for degree or professional qualifications were renamed as “Qualification 1”, “Qualification 2”, and so on.

To capture the start and end dates of each qualification, we renamed the columns as “From(Q1)”, “From(Q2)”, etc., for the starting dates, and “To(Q1)”, “To(Q2)”, etc., for the ending dates. The columns for the class associated with the qualifications were renamed as “Grade(Q1)”, “Grade(Q2)”, and so on.

A similar renaming process was applied to the experience data. Columns for company names were renamed as “Company 1”, “Company 2”, and so forth, while positions held were renamed as “Position 1”, “Position 2”, and so on. The start and end dates for job positions were renamed as “From(P1)”, “From(P2)”, etc., for the start dates, and “To(P1)”, “To(P2)”, etc., for the end dates. This renaming ensured that the data was well-organized and easily interpretable.

Text Format Standardization (Change Case)

To ensure consistency in presentation, we reformatted all text entries into sentence case. This change applied to institute names, degree names, company names, and job positions, making sure that all names followed standard capitalization rules. For instance, entries like "university of Moratuwa" were corrected to "University Of Moratuwa". This not only improved readability but also made the data easier to compare across entries.

Renaming Null Values

For all columns that did not contain date information (e.g., institute names, degrees, company names, and positions), we replaced any null values with the label "Undefined". This ensured that the dataset had no empty or ambiguous fields, making it easier to analyze.

Removing Duplicates

From the dataset, there was only one duplicate value noted. To remove it, we selected all the columns except for the ID and used the option to remove duplicate rows, which eliminated that value.

Standardization of Institute and University Names

To maintain consistency and avoid variation in institute and university names, we applied a standardization process. For example, all variations of AAT (such as Associate of Accounting Technicians) were replaced with the standardized form “AAT Sri Lanka”. Similarly, any values similar to CMA (such as Certified Management Accounting) were unified as “CMA”.

The names similar to CIMA (e.g., Chartered Institute of Management Accountants) was standardized as “CIMA”, while values similar to CIPM (e.g., Chartered Institute of Personnel Management Sri Lanka) was replaced with “CIPM”. For institutions like the “Institute of Bankers of Sri Lanka”, we replaced any similar variations with the official title. The name “CA Sri Lanka” was used to represent the Institute of Chartered Accountants of Sri Lanka consistently across the data.

For university names, entries like University of Moratuwa, University of Sri Jayewardenepura, and General Sir John Kotelawala Defence University were standardized to reflect their correct full names, avoiding abbreviations or misspellings.

Standardization of Degree and Professional Qualification Names

In addition to institute names, we also standardized the names of degrees and professional qualifications. For example, qualifications similar to AAT (such as Diploma in Accounting & Business and Financial Accounting - Professional Qualification) were replaced with “Diploma in Accounting & Finance”. This provided uniformity in how qualifications were listed.

Similarly, variations of CIMA (e.g., "CIMA", "CIMA Sri Lanka") were replaced with “CGMA Professional Qualification”. This ensured that candidates' qualifications were clearly categorized and presented consistently.

Classifications Using Column Tools

We used the grouping feature in the column tools tab of the table view to classify certain columns, adding an extra layer of organization to the data.

Degree Classification

To organize the grades associated with the candidates' qualifications, we focused on the Grades(Q1) and Grades(Q2) columns, as only Qualification 1 and Qualification 2 contained degree-related qualifications. Each of these grade columns was classified into standard degree categories: **First Class**, **Second Class Upper**, **Second-Class Lower**, and **General Degree**. Any entries that did not fit into these categories, as well as missing or unclear values, were labeled as **Undefined**. After the classification, the columns were renamed as “**Degree Classification 1**” and “**Degree Classification 2**”.

Institute Classification

We focused on classifying the Institute 1 column because many rows contained the same university names, which required standardization. Recognized universities and institutions, such as the **University of Moratuwa**, **University of Sri Jayewardenepura**, and **General Sir John Kotelawala Defence University**, were grouped accordingly. Any entries that did not match these recognized institutions, or were ambiguous, were labeled as **Undefined**. After the classification, the columns were renamed as “**Institute 1(Group).**”

Qualification Duration Calculation

To calculate the duration of each qualification, we considered the From and To dates for each qualification and computed the duration in years. For instance, to get the duration for the first qualification, we used the DAX formula:

```
Duration(Q1) = DATEDIFF('Applicant Dataset for Visualization'[From(Q1)], 'Applicant Dataset for Visualization'[To(Q1)], YEAR)
```

This DAX formula calculates the difference between the start date (From(Q1)) and the end date (To(Q1)) in years. The resulting duration was then stored in a new column, which was renamed **“Duration(Q1)”**.

We followed a similar procedure to calculate the duration for the second, third, and subsequent qualifications, renaming the columns as **“Duration(Q2)”**, **“Duration(Q3)”**, and so on.

Eligibility Category Classification

The classification of candidates' qualifications was carried out through a detailed multi-step process that focused on the Qualification 1, Duration(Q1), and Institute 1 columns. The first step involved determining if Qualification 1 contained the term "Hons" (indicating an honors degree) and whether the Duration(Q1) was 4 years or more. If both conditions were met, the qualification was classified as an **“Honors Degree.”** In cases where the duration was also 4 years or longer but did not include "Hons," the qualification was categorized as a **“4-Year Degree”**.

Next, qualifications associated with Institute 1 were examined. If the institute was identified as **CIMA**, the qualification was classified as a **“3-Year Degree”**. Similarly, if the Duration(Q1) was exactly 3 years, it was also classified as a **“3-Year Degree”**. For professional qualifications, if Institute 1 matched well-known institutions like **CA**, **CIM**, or **CIPM**, the qualification was classified as **“Professional Qualifications”**. Any entry labeled as "Undefined" or missing was categorized as **“Undefined”**. Finally, qualifications that did not fit into the above categories (such as Masters degrees) were classified as **“Other”**.

This is the DAX formula used for classifying the Qualification 1:


```

Eligibility Category(Q1) =
    IF(
        CONTAINSSTRING([Qualification 1], "Undefined"),
        "Undefined",
        IF(
            CONTAINSSTRING([Institute 1], "CIMA"),
            "3-Year Degree",
            IF(
                CONTAINSSTRING([Institute 1], "CA") ||
                CONTAINSSTRING([Institute 1], "CIM") ||
                CONTAINSSTRING([Institute 1], "CIPM"),
                "Professional Qualifications",
                IF(
                    CONTAINSSTRING([Qualification 1], "Hon") && [Duration(Q1)] >= 4,
                    "Honors Degree",
                    IF(
                        [Duration(Q1)] >= 4,
                        "4-Year Degree",
                        IF(
                            [Duration(Q1)] = 3,
                            "3-Year Degree",
                            "Other"
                        )
                    )
                )
            )
        )
    )

```

After applying these classification rules, the resulting column was renamed “**Eligibility Category(Q1)**” to reflect the categorization of the Qualification 1. This classification procedure was consistently applied to all subsequent qualifications, leading to the creation of columns like “**Eligibility Category(Q2)**”, “**Eligibility Category(Q3)**”, and so on.

Effective Date Calculation

To determine the completed date of the first qualification for each candidate, we analyzed all the **To(Q1)**, **To(Q2)**, **To(Q3)**, and **To(Q4)** columns. The goal was to identify the earliest completed date among these qualifications and store it in a new column named “**Effective Date**”

This is the DAX formula used:

```

Effective Date =
MINX(
    {
        [To(Q1)],
        [To(Q2)],
        [To(Q3)], [To(Q4)]
    },
    [Value]
)

```

- In this DAX formula, the **MINX** function was used to evaluate the minimum date value across the specified From columns.

Position Duration Calculation

To determine the duration of each position held by candidates, we focused on the From and To dates associated with each position. The duration was calculated in years and recorded in new columns labeled “**PositionDuration(P1)**”, “**PositionDuration(P2)**”, etc.

The process began by checking whether the From(P1) date was greater than or equal to the Effective Date and less than or equal to the **advertisement date of September 30, 2024**. If both conditions were true, we calculated the duration using the **DATEDIFF** function, which measures the time between From(P1) and To(P1). If To(P1) was blank, we used September 30, 2024, as the end date. If the conditions were not met, the duration was set to 0.

This is the DAX formula used for calculating the duration of the first position:

```

PositionDuration(P1) =
IF (
    AND (
        [From(P1)] >= [Effective Date],
        [From(P1)] <= DATE(2024, 9, 30)
    ),
    DATEDIFF(
        [From(P1)],
        IF ( ISBLANK([To(P1)]), DATE(2024, 9, 30), [To(P1)] ),
        YEAR
    ),
    0
)

```

Total Position Duration Calculation (Calculating Post Qualifying Experience)

To calculate the total position duration for each candidate, we summed the individual durations recorded in the **PositionDuration** columns. The new column created for this total was labeled **“PostQualifyingExperience”**.

This is DAX formula used for this calculation:

```

PostQualifyingExperience = [PositionDuration(P1)] + [PositionDuration(P2)] + [PositionDuration(P3)]
                           + [PositionDuration(P4)] + [PositionDuration(P5)] +
                           + [PositionDuration(P6)] + [PositionDuration(P7)]

```

Qualification Classification according to degree type

We used a grouping method to determine the degree type for each candidate by classifying them based on their qualifications.

First, if a candidate's degree was in Business/Management or a closely related field, it was classified as **Business/Management**. Conversely, degrees categorized as Non-business/nonmanagement were classified as **Non Business**. For professional qualifications, such as AAT, CIMA, and CA, these were grouped under the category of **Professional Qualification**. Additionally, any qualifications that did not qualify as degrees (such as diplomas and National Diplomas in Engineering Studies (NDES)), were labeled as **Non-Degree**. Lastly, qualifications marked as Undefined were classified as **Undefined**. This classification process was consistently

applied to other qualifications. After classification, the columns were renamed as “**Classification 1**”, “**Classification 2**”, and so on, corresponding to each qualification.

Institute Location Classification

We classified the institutes based on their geographic location. This process focused on the Institute 1, Institute 2, Institute 3, and Institute 4 columns. Each institute was categorized by its city and country. For example, the **University of Uva Wellassa** was classified as **Badulla, Sri Lanka**, and the **Dalian Maritime University** was classified as **Dalian, China**.

After completing the classification, the columns were renamed as “**Location (Institute 1)**”, “**Location (Institute 2)**”, and so on, providing a clear indication of each institution's geographic location.

Eligibility Criteria for Senior Managers and Entrepreneurs

In the eligibility criteria, we specifically looked for candidates who fell into the category of Senior Managers and Entrepreneurs with an exceptional track record. To identify such candidates, we focused on the Position columns, beginning with Position 1. We implemented a DAX formula to flag any relevant titles in this column as follows:

```
SpecialTrack(P1) =  
SWITCH(  
    TRUE(),  
    -- Flag as "Senior Manager" if the position contains titles like CEO, Managing Director, etc.  
    CONTAINSSTRING([Position 1], "CEO") ||  
    CONTAINSSTRING([Position 1], "Managing Director") ||  
    CONTAINSSTRING([Position 1], "General Manager") ||  
    CONTAINSSTRING([Position 1], "Operations Director") ||  
    CONTAINSSTRING([Position 1], "Business Development Director") ||  
    CONTAINSSTRING([Position 1], "Strategic Advisor") ||  
    CONTAINSSTRING([Position 1], "Project Director"),  
    "Senior Manager",  
    -- Flag as "Entrepreneur" if the position contains titles like Founder, Co-founder, or Entrepreneur  
    CONTAINSSTRING([Position 1], "Founder") ||  
    CONTAINSSTRING([Position 1], "Co-founder") ||  
    CONTAINSSTRING([Position 1], "Entrepreneur"),  
    "Entrepreneur",  
    -- Default if none of the above apply  
    BLANK()  
)
```

We applied this same procedure to all subsequent **Position** columns. If the resulting columns were entirely blank after classification, we removed them from the dataset. Ultimately, only the

“**SpecialTrack(P1)**” column remained, which identified the candidate with a senior management title.

Eligibility Criteria

To determine the eligibility of candidates based on their qualifications and professional experience, we implemented a comprehensive classification system. The process begins by evaluating candidates with an Honors Bachelor's degree in Business from the University of Moratuwa. If they possess this qualification, they are categorized as having "**Direct Eligibility(category 1)**" Next, candidates with a four-year degree in Business or a related field are also classified under "**Direct Eligibility(category 2)**" For candidates holding a four-year degree in a non-business area, they must have at least one year of relevant work experience to qualify as "**Eligible with Experience(category 3)**"

Additionally, candidates with a three-year degree, whether in business or a non-business discipline, can be eligible if they possess two or more years of relevant experience, classified as "**Eligible with Experience(category 4)**" Recognized professional qualifications, such as CIMA, can also qualify candidates if they have a minimum of two years of experience, leading to the category "**Eligible(category 5)**" Furthermore, a “**Special Admission(category 6)**” category is available for Senior Managers and Entrepreneurs, contingent upon their titles in the Position columns, but this is limited to 10% of the total class size. If none of these conditions are met, the candidate is deemed "**Not Eligible.**" This column was renamed as “**Eligibility Criteria**”.

Here is the DAX formula that implements this eligibility classification:

```

Eligibility Criteria =
SWITCH(
    TRUE(),

-- Direct eligibility for Honors Bachelor's Degree in Business from University of Moratuwa
    (CONTAINSSTRING([Qualification 1], "Hon") && CONTAINSSTRING([Qualification 1], "Business") && CONTAINSSTRING([Institute 1], "University
of Moratuwa")) ||
    (CONTAINSSTRING([Qualification 2], "Hon") && CONTAINSSTRING([Qualification 2], "Business") && CONTAINSSTRING([Institute 2], "University
of
Moratuwa")) ||

    (CONTAINSSTRING([Qualification 3], "Hon") && CONTAINSSTRING([Qualification 3], "Business") && CONTAINSSTRING([Institute 3], "University
of Moratuwa")) ||
    (CONTAINSSTRING([Qualification 4], "Hon") && CONTAINSSTRING([Qualification 4], "Business") && CONTAINSSTRING([Institute 4], "University
of Moratuwa")) ||
    "Direct Eligibility(category 1)",

-- Direct eligibility for Four-year Degree or Honors Degree in Business/Management or related area
    ((CONTAINSSTRING([Eligibility Category(Q1)], "4-Year Degree") || CONTAINSSTRING([Eligibility Category(Q1)], "Honors Degree")) &&
CONTAINSSTRING([Classification 1], "Business/Management")) ||
    ((CONTAINSSTRING([Eligibility Category(Q2)], "4-Year Degree") || CONTAINSSTRING([Eligibility Category(Q2)], "Honors Degree")) &&
CONTAINSSTRING([Classification 2], "Business/Management")) ||
    ((CONTAINSSTRING([Eligibility Category(Q3)], "4-Year Degree") || CONTAINSSTRING([Eligibility Category(Q3)], "Honors Degree")) &&
CONTAINSSTRING([Classification 3], "Business/Management")) ||
    ((CONTAINSSTRING([Eligibility Category(Q4)], "4-Year Degree") || CONTAINSSTRING([Eligibility Category(Q4)], "Honors Degree")) &&
CONTAINSSTRING([Classification 4], "Business/Management")) ||
    "Direct Eligibility(category 2)",

-- Eligibility with experience for Four-year Non-business/non-management degree or Honors Degree with 1+ years of experience
    ((CONTAINSSTRING([Eligibility Category(Q1)], "4-Year Degree") || CONTAINSSTRING([Eligibility Category(Q1)], "Honors Degree")) &&
CONTAINSSTRING([Classification 1], "Non Business") && [PostQualifyingExperience] >= 1) ||
    ((CONTAINSSTRING([Eligibility Category(Q2)], "4-Year Degree") || CONTAINSSTRING([Eligibility Category(Q2)], "Honors Degree")) &&
CONTAINSSTRING([Classification 2], "Non Business") && [PostQualifyingExperience] >= 1) ||
    ((CONTAINSSTRING([Eligibility Category(Q3)], "4-Year Degree") || CONTAINSSTRING([Eligibility Category(Q3)], "Honors Degree")) &&
CONTAINSSTRING([Classification 3], "Non Business") && [PostQualifyingExperience] >= 1) ||
    ((CONTAINSSTRING([Eligibility Category(Q4)], "4-Year Degree") || CONTAINSSTRING([Eligibility Category(Q4)], "Honors Degree")) &&
CONTAINSSTRING([Classification 4], "Non Business") && [PostQualifyingExperience] >= 1) ||
    "Eligible with Experience(category 3)",

-- Eligibility with experience for Three-year Degree with 2+ years of experience
    (
        ((([Classification 1] = "Business/Management" || [Classification 1] = "Non Business") && CONTAINSSTRING([Eligibility Category(Q1)],
"3-Year Degree") && [PostQualifyingExperience] >= 2) ||
        ([Classification 2] = "Business/Management" || [Classification 2] = "Non Business") && CONTAINSSTRING([Eligibility Category(Q2)],
"3-Year Degree") && [PostQualifyingExperience] >= 2) ||
        ([Classification 3] = "Business/Management" || [Classification 3] = "Non Business") && CONTAINSSTRING([Eligibility Category(Q3)],
"3-Year Degree") && [PostQualifyingExperience] >= 2) ||
        ([Classification 4] = "Business/Management" || [Classification 4] = "Non Business") && CONTAINSSTRING([Eligibility Category(Q4)],
"3-Year Degree") && [PostQualifyingExperience] >= 2) ||
    ),
    "Eligible with Experience(category 4)",

-- Eligibility for Recognized Professional Qualification (e.g., CIMA) with 2+ years of experience
    ([Classification 1] = "Professional Qualification" && [PostQualifyingExperience] >= 2) ||
    ([Classification 2] = "Professional Qualification" && [PostQualifyingExperience] >= 2) ||
    ([Classification 3] = "Professional Qualification" && [PostQualifyingExperience] >= 2) ||
    ([Classification 4] = "Professional Qualification" && [PostQualifyingExperience] >= 2),
    "Eligible(category 5)",

-- Special admission for Senior Managers and Entrepreneurs (manual input or additional criteria check)
    IF(
        CONTAINSSTRING([SpecialTrack(P1)], "Senior Manager") ||
        CONTAINSSTRING([SpecialTrack(P1)], "Entrepreneur"),

        -- Limiting to 10% of the class size
        IF(
            COUNT('Applicant Dataset for Visualization'[SpecialTrack(P1)]) <=6,
            "Special Admission(category 6)",
            "Not Eligible (Exceeds 10% Limit)"
        ),

        -- If none of the above conditions are met
        "Not Eligible"
    )
)

```

Eligibility Classification

To categorize the candidates as either eligible or not eligible, we classified the Eligibility Criteria column. Candidates who met the program requirements were categorized under Eligible, with specific categories such as Direct Eligibility(Category 1), Direct Eligibility(Category 2), and so on, covering all six eligibility categories. Those who did not meet the requirements were categorized as Not Eligible. After this classification, the column was renamed **“Eligibility”** to clearly reflect the candidates' eligibility status.

Dashboard Design with key visuals & advanced features

After completing the data preparation, cleaning, and data wrangling processes, we moved on to the visualization stage. At this point, we focused on designing the dashboard, titled **"UoM Master's Eligibility Summary"** which consists of three pages. The dashboard incorporates the functional requirements, key visuals, and advanced features to effectively present the data. The visualizations were carefully crafted to ensure clarity and provide meaningful insights from the dataset.

Eligibility Criteria Slicer

There are several categories to assess candidates for eligibility. Candidates with an Honors Bachelor's degree in Business from the University of Moratuwa are classified as having "Direct Eligibility(category 1)" Similarly, candidates with a four-year degree in Business or a related field also fall under "Direct Eligibility(category 2)"

For those holding a four-year degree in a non-business discipline, at least one year of relevant work experience is required to be classified as "Eligible with Experience(category 3)" Candidates with a three-year degree, whether in business or a non-business field, must have two or more years of relevant experience to qualify as "Eligible with Experience(category 4)"

Recognized professional qualifications, such as CIMA, qualify candidates under the "Eligible(category 5)" category if they have at least two years of experience. Additionally, a "Special Admission(category 6)" category is available for Senior Managers and Entrepreneurs, based on their titles in the Position columns, but this category is capped at 10% of the total class size. If none of these conditions are met, the candidate is classified as "Not Eligible."

To represent these categories, we created a slicer using the Eligibility Criteria column and titled it as "Eligibility Criteria" to allow users to easily filter and analyze candidates based on their eligibility status, ensuring a more dynamic and interactive view of the dataset.

Degree Classification Slicer

To display the degree classification (such as First Class, Second Class Upper, Second Class Lower, and General Degree) for each candidate, we created a slicer using the Degree Classification 1 column. This allows users to filter the data based on the degree categories. The slicer was labeled as “Degree Classification” to facilitate easy selection and analysis of the degree types across candidates. Eligibility Category Slicer

To represent the eligibility category for each candidate (such as 3-Year Degree, 4-Year Degree, Honors Degree, Other, and Professional Qualifications), we created a slicer using only the EligibilityCategory(Q1) column. This allows users to filter candidates based on their primary eligibility category. The slicer was labeled as "**Eligibility Category**" to provide a clear view of the candidates' qualifications and eligibility status, making it easier to navigate and assess individual eligibility.

Post Qualifying Experience Slicer

To represent the post-qualifying experience of each candidate, we created a slicer using the PostQualifyingExperience column. This slicer allows users to filter candidates based on their total years of experience by adjusting the starting and ending time of the duration. The slicer was renamed as “Post Qualifying Experience” to clearly reflect the candidates' professional experience after their qualifications.

Academic Institutions Slicer

To represent the academic institutions of the candidates, we created a slicer using only the Institute 1 (group) column. The slicer was renamed "**Academic Institution**" and displayed as a dropdown list, allowing users to filter candidates based on the primary university or institution they attended. This simplifies the filtering process and provides a clear view of candidates' academic backgrounds.

Applicant Information (Table)

To represent the applicant information, we created a table titled "**Applicant Information**" using the following columns: id, Institute 1(group), Qualification 1, Institute 2, Qualification 2, PostQualifyingExperience, and Eligibility Criteria. This table provides a detailed overview of each candidate's academic background, qualifications, post-qualifying experience, and their eligibility status, offering a comprehensive view of the applicant pool for easy analysis.

Geographic Distribution of Institutes (Map Visualization)

To show the geographic distribution of candidates based on their academic institutions, we used a map visualization. Before creating the map, we set the data category (in the column tools properties) as "Place" for all the location columns (e.g., Location(Institute 1), Location(Institute 2), and so on). After configuring the data, we used these columns to plot the institutions on the map and renamed the visual as "**Location Distribution of Institutes**". We also included zoom buttons and displayed category labels to enhance visibility and facilitate better navigation of the map. This visual representation helps in identifying the geographical spread of the candidates' academic backgrounds.

Distribution of Academic Institutions (Tree Map Visualization)

To represent the distribution of academic institutions, we utilized a tree map visualization, titled "**Classification according to Institution**". In this visual, we used the count of IDs as the values

and the institutes (Institute 1, Institute 2, and so on) as the categories. This approach allows for a clear comparison of the number of candidates associated with each institution, providing insights into which academic institutions have the highest representation among the candidates. The tree map effectively visualizes the relative sizes of each institution's contribution to the total candidate pool. Users can interact with the tree map to explore the distribution separately for each institution while also having the option to view the overall distribution.

Degree Type Classification Funnel Chart

To visually represent the degree classifications, we utilized a funnel chart titled as the **“Classification According to Degree Type”**. In the classification process, candidates' degrees in Business/Management or related fields were grouped under Business/Management, while degrees outside these fields were classified as Non Business. Professional qualifications like AAT, CIMA, and CA were categorized under Professional Qualification, while non-degree qualifications (such as diplomas or NDES) were labelled as Non-Degree. Any qualifications that were unclear or not defined were categorized as Undefined. The funnel chart used the count of IDs for values and Classification 1, Classification 2, and so on for categories. This funnel chart enables users to explore the distribution of candidates across the various classifications (Classification 1, Classification 2) separately, while also providing the option to view an overall summary of degree classifications.

Eligibility Criteria Distribution (Line and Stack Column Chart Visualization)

To represent the eligibility criteria of candidates, we utilized a line and stack column chart titled "Count of id by Eligibility Criteria" In this visual, the x-axis represents the various eligibility criteria, such as Direct Eligibility(category 1),Direct Eligibility(category 2) and so on. The y-axis shows the count of candidate IDs, providing a clear view of the distribution of students across different eligibility categories. This approach allows for easy comparison of the number of candidates in each eligibility category, offering valuable insights into the qualifications of the candidates.

Payment Verification (Pie Chart)

To visually represent whether candidates have paid their fees, we used a pie chart titled **“Count of applicants by pay_verified”**. The chart utilized the pay_verified column, where a value of 1 indicated verified payment and 0 indicated non-payment. The count of candidate IDs was used for the values, providing a clear view of how many candidates had completed the payment process versus those who had not.

Eligibility Category Distribution (Donut Chart)

To represent the distribution of eligibility categories, such as 4-Year Degree and 3-Year Degree, we utilized a donut chart titled **“Count of applicants by Eligibility Category”**. In this visual, the count of candidate IDs was used as the values, while the eligibility categories from the columns EligibilityCategory(Q1), EligibilityCategory(Q2), and so on were given as legends in the chart. This approach allows us to examine the distribution of eligibility categories for each qualification separately, and users can navigate through the legends to explore each category in detail. Additionally, an overall representation of the eligibility categories is also provided, enhancing the understanding of candidates' qualifications.

Gender Classification Stacked Bar Chart

To represent the gender classification of the candidates, we created a stacked bar chart using the count of ID and the gender column. This visual breakdown displays one bar for male candidates and another bar for female candidates in the dataset. The chart was labeled as **“Gender Classification”** to clearly show the distribution of candidates by gender.

First Qualification Date Distribution (Stacked Bar Chart Visualization)

To visually represent the distribution of the first qualification dates among candidates, we used a stacked bar chart. The Effective Date refers to the earliest completed qualification date for each candidate, based on their academic history. In the chart, the Y-axis displayed this Effective Date

broken down by Year, Quarter, Month, and Day, while the X-axis represented the count of IDs. This approach allows users to expand the visualization level-wise, enabling a more detailed examination of when candidates completed their first qualifications over time. Users can drill down from Year to Quarter, Month, and Day, providing a comprehensive view of the timeline of candidates' qualifications. The chart was titled "**Distribution of First Qualification**" to reflect its purpose of showcasing this timeline.

Distribution of Companies (Tree Map Visualization)

To represent the distribution of companies where candidates have worked, we utilized a tree map visualization, titled "**Count of applicants by Companies**". In this visual, we used the count of IDs as the values and the companies (Company 1, Company 2, and so on) as the categories. This approach allows for a clear comparison of the number of candidates associated with each company, offering insights into which organizations have the highest representation among the candidates. The tree map effectively visualizes the relative sizes of each company's contribution to the total candidate pool. Users can interact with the tree map to explore the distribution for each company separately, while also having the option to view the overall distribution.

Distribution of Post Qualifying Experience (Donut Chart Visualization)

To represent the distribution of post qualifying experience, we utilized a donut chart visualization titled "**Count of Applicants by Post qualifying experience**". In this chart, the count of IDs was used as the values, while the PostQualifyingExperience was represented as legends. This visualization provides a clear breakdown of candidates' work experience after qualifying, showing the proportion of candidates with different lengths of professional experience.

Eligibility Distribution (Funnel Chart Visualization)

To represent the applicants by eligibility, we used a funnel chart. In this visual, we employed the Eligibility column as the category and the count of IDs as the values. The chart effectively displays

the number of candidates who are eligible and those who are not eligible. The chart was titled **"Count of Applicants as Eligibility"** to reflect its purpose in showcasing the distribution of applicants based on their eligibility status.

Export Power Automated Button:

We can use this feature to export a CSV file to an Excel file on OneDrive. We have developed this based on the video provided in the references. This allows us to obtain the filtered values that can be used for future analysis.

Conclusion

In conclusion, this assignment underscores the importance of structured data analysis and visualization in the admissions process for the Master of Business Analytics program. Through the creation of the Power BI dashboard, we have successfully categorized and visualized candidate data based on their academic qualifications, professional experiences, and eligibility criteria as outlined in the assignment briefing. The interactive nature of the dashboard, enhanced by various visualization tools, not only aids in understanding candidate distributions but also facilitates strategic decision-making for admissions committees. With features like eligibility filters, visual indicators, and the ability to calculate experience automatically, the dashboard serves as a comprehensive resource for effectively navigating the complexities of candidate assessment and selection. As the demand for data-driven insights continues to grow, the methods and visualizations presented in this dashboard will be invaluable for educational institutions aiming to streamline their admissions processes.

Please find the published document of power bi dashboard:(Group 6)

<https://app.powerbi.com/links/tiW9PqzPFZ?ctid=aac0c564-6c5e-4b05-8dc3->

[408087f77f76&pbi_source=linkShare](#)

References

- <https://www.youtube.com/watch?si=xxcbR6leWs9LHXKu&v=sd3YCjN8Co4&feature=youtu.be>
- <https://youtu.be/x260sd1KsqE?si=ts4QyOtwPyk-Djcr>